

Fashion Recommendation System

Using Collaborative Filtering and Cosine Similarity

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Business Problem

- Fashion platforms contain thousands of products, making product discovery difficult for users.
- Generic recommendations reduce user engagement and personalization.
- Businesses require intelligent recommendation systems to improve customer experience and increase sales.

Project Objective

- Analyze fashion product and user rating data
- Build recommendation models for personalized suggestions
- Implement collaborative filtering using cosine similarity
- Compare recommendation approaches
- Visualize insights using Tableau dashboards
- Deploy the recommendation system using Streamlit

Dataset Overview

Feature	Description
User ID	Unique user identifier
Product ID	Unique product identifier
Product Name	Fashion product
Brand	Product brand
Category	Product category
Rating	User rating
Color	Product color
Size	Product size
Price	Product price

Dataset consists of user-product interactions used for collaborative filtering recommendations.

Data Preprocessing

- Encoded categorical variables using LabelEncoder
- Created a user-item interaction matrix
- Handled missing values using zero imputation
- Built a preprocessing pipeline for workflow consistency
- Prepared data for similarity computation

Baseline vs Advanced Model

Baseline Model

Popularity-Based Recommendation

- Recommends highest-rated products overall
- Fast and computationally efficient
- Works for new users
- Not personalized

Advanced Model

Collaborative Filtering

- Uses cosine similarity
- Identifies similar users
- Generates personalized recommendations
- More personalized recommendations

Recommendation Workflow

Dataset



Preprocessing



User-Item Matrix



Cosine Similarity



Collaborative Filtering



Recommendations



Evaluation & Visualization



Streamlit + Tableau

Evaluation Metrics

- Collaborative filtering was used to recommend products based on similar user preferences.
- Recommendation relevance was evaluated qualitatively
- Popularity-based fallback improved cold-start handling
- Similarity scores
- Lower values indicate better prediction accuracy

Tableau Dashboard

An interactive Tableau dashboard was created to visualize:

- Distribution of products by category
- Average ratings across brands
- Product rating distributions

The dashboard helps stakeholders understand:

- Product trends
- Customer preferences
- Brand performance

- Kid's Fashion had the highest product count.
- Gucci products showed the highest average ratings.
- Most product ratings were concentrated between 2.5 and 4.5.

Streamlit Web Application

A simple interactive web application was developed where:

- Users enter a User ID
- The system generates personalized recommendations

The Streamlit application provides an interactive demonstration of personalized recommendations.

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Key Insights

- Users with similar rating behaviors tend to prefer similar fashion products.
- Collaborative filtering improved personalization compared to popularity-based recommendations.
- Popularity-based recommendations provided stable fallback suggestions for new users.

Challenges Encountered

- Sparse user-item interaction matrix
- Cold-start problem for new users
- Limited recommendation diversity
- Synthetic dataset limitations

Conclusion & Future Improvements

Conclusion

Collaborative filtering successfully generated personalized fashion product recommendations using user similarity patterns.

Future improvements could include:

- Hybrid recommendation systems
- Deep learning recommenders
- Real-time recommendations
- Integration with real-world e-commerce data

Thank You



Thank You